



REVIEW-II

LexAgent

Multi-Agent Debate Framework for Hallucination Reduction in Legal AI

Student Name(s)

23LC1242 – Rishi Jain
23BLC1115 – Lakshya Agarwal
23BEC1242 – Aditya Pratap Singh Takuli

Programme / School

B.Tech, SENSE, VIT Chennai

Guide

Dr. Raju Patel [53066]

Review & Date

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GUIDE'S SIGNATURE WITH DATE

Signature: _____

Date: _____

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Detailed Literature Review

Papers 1–5 of 27 · peer-reviewed



NO.	AUTHOR(S) & YEAR	TITLE / SOURCE	METHOD / APPROACH	FINDINGS & RESEARCH GAP
[1]	Xiao et al., 2021	Lawformer: pre-trained LM for Chinese legal long documents · AI Open	Longformer-style pre-training on long Chinese legal documents	Strong long-document legal understanding; no retrieval or citation checking
[2]	Lewis et al., 2020	Retrieval-augmented generation for knowledge-intensive NLP · NeurIPS	Dense retriever + seq2seq generator trained end-to-end	Retrieval grounds generation; no check that sources support the answer
[3]	Chalkidis et al., 2022	LexGLUE: legal language understanding in English · ACL	Benchmark of seven English legal NLU tasks	Standard legal NLU scoring; no citation or hallucination metric
[4]	Wang, X. et al., 2023	Self-consistency improves chain-of-thought reasoning · ICLR	Sample several reasoning paths; majority vote	Consistency lifts accuracy; consistent answers can still lack evidence
[5]	Smit et al., 2024	Should we be going MAD? Multi-agent debate strategies for LLMs · ICML	Benchmarks debate strategies against prompting baselines	Debate gains are configuration-sensitive; no legal citation matcher

Detailed Literature Review (contd.)

Papers 6–10 of 27



NO.	AUTHOR(S) & YEAR	TITLE / SOURCE	METHOD / APPROACH	FINDINGS & RESEARCH GAP
[6]	Dahl et al., 2024	Large legal fictions: profiling legal hallucinations in LLMs · J. Legal Analysis	Verifiable U.S. federal case questions posed to public LLMs	58–88% hallucination; motivates citation-level verification
[7]	Fei et al., 2024	LawBench: benchmarking legal knowledge of LLMs · EMNLP	Chinese legal benchmark: memorise, understand, apply	Broad legal scoring; no measure of citation refusal
[8]	Anaokar et al., 2025	HalluDetect: hallucinations in legal conversational systems · EMNLP Industry	Hallucination detection for legal chat systems	Post-hoc detection; no memory of past fabrications
[9]	Liu & Tang, 2025	LeAK-GraphRAG: legal academic knowledge GraphRAG · JURIX	Knowledge-graph RAG over legal academic sources	Better-structured evidence; no adversarial check of claims
[10]	Huang et al., 2026	Mitigating legal hallucinations via symbolic constraints and analogical precedents (AALawyer) · ACL	Symbolic constraints with analogical precedent retrieval	Constrains generation; no independent citation veto

Detailed Literature Review (contd.)

Papers 11–15 of 27



NO.	AUTHOR(S) & YEAR	TITLE / SOURCE	METHOD / APPROACH	FINDINGS & RESEARCH GAP
[11]	Cui et al., 2026	ChatLaw: multi-agent legal assistant with role-aligned MoE · Fundamental Research	Role-aligned mixture-of-experts legal agents	Role specialisation helps; agents share one evidence pool
[12]	Noël et al., 2026	HalluGraph: auditable hallucination detection for legal RAG · AAAI	Graph alignment of entities and relations	Entity and path checks beat embedding similarity; post-hoc audit only
[13]	Bose, 2026	Falkor-IRAC: graph-constrained generation for Indian judicial AI · InIRAC dataset	Graph-constrained IRAC-structured generation	Reduces drift; jurisdiction-specific, no debate
[14]	Meng et al., 2025 (base paper)	Multi-agent human-in-the-loop framework for legal terminology mapping · AI and Law	Role-specialised agents with human review	Motivates LexAgent's role decomposition; no citation verification
[15]	Hu et al., 2025	Removal of hallucination on hallucination: Debate-Augmented RAG · ACL	Debate at both retrieval and generation stages	Reduces compounding hallucination; general domain, no legal citation check

Detailed Literature Review (contd.)

Papers 16–21 of 27



NO.	AUTHOR(S) & YEAR	TITLE / SOURCE	METHOD / APPROACH	FINDINGS & RESEARCH GAP
[16]	Du et al., 2023	Improving factuality and reasoning through multiagent debate · arXiv:2305.14325	Several LLM instances debate over rounds	Improves factuality; shared context risks false consensus
[17]	Liang et al., 2024	Encouraging divergent thinking through multi-agent debate · arXiv:2305.19118	Affirmative and negative debaters with a judge	Counters degeneration of thought; no evidence independence
[18]	Guha et al., 2023	LegalBench: measuring legal reasoning in LLMs · arXiv:2308.11462	Collaboratively built legal-reasoning task suite	Standard reasoning scores; no citation-hallucination metric
[19]	Yue et al., 2023	DISC-LawLLM: fine-tuning LLMs for intelligent legal services · arXiv:2309.11325	Legal instruction tuning plus a retrieval module	Legal fluency with retrieval; single generator, no citation veto
[20]	Dhuliawala et al., 2023	Chain-of-Verification reduces hallucination in LLMs · arXiv:2309.11495	Draft, plan verification questions, answer, revise	Fewer hallucinations; self-checks without external legal sources
[21]	Asai et al., 2023	Self-RAG: retrieve, generate and critique through self-reflection · arXiv:2310.11511	Reflection tokens trigger retrieval and critique	Adaptive retrieval; critique comes from the same model

Detailed Literature Review (contd.)

Papers 22–27 of 27



NO.	AUTHOR(S) & YEAR	TITLE / SOURCE	METHOD / APPROACH	FINDINGS & RESEARCH GAP
[22]	Yan et al., 2024	Corrective retrieval augmented generation · arXiv:2401.15884	Retrieval evaluator triggers corrective actions	Fixes poor retrieval; does not verify the written claim
[23]	Wang, Q. et al., 2024	Rethinking the bounds of LLM reasoning: are multi-agent discussions the key? · arXiv:2402.18272	Multi-agent discussion vs. single-agent prompts	Gains shrink with good prompts; motivates controlled ablations
[24]	Sun et al., 2024	Detecting LLM hallucination via Markov chain-based multi-agent debate · arXiv:2406.03075	Claim-level debate modelled as a Markov chain	Better detection; no legal citation matching
[25]	Zhou et al., 2024	LawGPT: a Chinese legal knowledge-enhanced LLM · arXiv:2406.04614	Legal-knowledge pre-training and instruction tuning	Domain fluency; no independent citation check
[26]	Li et al., 2024	LegalAgentBench: evaluating LLM agents in the legal domain · arXiv:2412.17259	Tool-using agent benchmark on legal tasks	Evaluates agents; no re-hallucination measure
[27]	Chowdhury et al., 2026	Courtroom-style multi-agent debate with progressive RAG (PROCLAIM) · arXiv:2603.28488	Courtroom roles with progressive retrieval	Adversarial roles and staged evidence; no citation verifier

Detailed Literature Review (contd.)

Literature Landscape: Five Method Families



FAMILY	PAPERS	CONTRIBUTION	GAP FOR LEGAL QA
Domain legal LLMs	Lawformer [1], DISC-LawLLM [19], LawGPT [25]	Legal fluency; DISC-LawLLM adds retrieval	Single generator; no independent citation veto
Legal benchmarks	LegalBench [18], LawBench [7], LexGLUE [3], LegalAgentBench [26]	Standard scoring of legal knowledge, reasoning and agents	No measure of citation refusal or re-hallucination
RAG and hybrid retrieval	RAG [2], CRAG [22], Self-RAG [21], LeAK-GraphRAG [9], AALawyer [10]	Grounds generation; corrects or constrains retrieval	Better evidence, but no adversarial check of the written claim
Detection and graph checks	Self-consistency [4], CoVe [20], HalluDetect [8], HalluGraph [12], Falkor-IRAC [13]	Entity and path checks beat embedding similarity	Post-hoc audit; no memory of past fabrications
Multi-agent debate	Du [16], MAD [17], Smit [5], Wang [23], Sun [24], ChatLaw [11], PROCLAIM [27]	Adversarial roles and progressive retrieval reduce false consensus	Configuration-sensitive; no legal citation matcher

Detailed Literature Review (contd.)



Comparison of the Closest Systems

SYSTEM · DOMAIN	ARCHITECTURE	ADVERSARIAL DEBATE	IN-LOOP CITATION CHECK	HYBRID / STRUCTURED RETRIEVAL	CROSS-SESSION MEMORY
ChatLaw [11] Chinese law	RA-MoE, cooperative SOP	☐ No	☐ Semantic entropy	☒ Retrieval + KG	☐ No
HalluGraph [12] Legal RAG	Graph-alignment auditor	☐ No	☐ Post-generation	☐ No	☐ No
HalluDetect [8] Indian consumer law	Detector + cooperative agents	☐ No	☐ Post-generation	☒ RAG chatbots	☐ No
LeAK-GraphRAG [9] Chinese legal papers	GraphRAG	☐ No	☐ No	☒ Vector + Cypher	☐ No
AALawyer [10] Chinese criminal law	Dual RAG (SCR + APR)	☐ No	☐ Constrained retrieval	☒ Symbolic + analogical	☐ No
Falkor-IRAC [13] Indian judgments	IRAC graph + verifier veto	☐ Single generator	☒ Veto after proposal	☒ Symbolic graph	☐ No
PROCLAIM [27] Biomedical claims	Courtroom MAD + P-RAG	☒ Yes · $\approx 11\times$ tokens	☐ Evidence, not citations	☒ P-RAG only	☐ No
LexAgent (ours) U.S. federal case law	4-agent debate on one T4 GPU	☒ Evidence-disjoint	☒ $CCS = E \times S$	☒ BM25 + Legal-BERT	☒ Verdict memory + risk

Detailed Literature Review (contd.)

Research Gaps Identified and LexAgent Response



GAP IN THE LITERATURE	LEXAGENT RESPONSE	VALIDATION STILL NEEDED
Retrieval can introduce misleading evidence [15, 21, 22]	Hybrid retrieval + reflection-driven query expansion	Recall of target cases and passages
Debate repeats shared errors [5, 17]	Defense counter-retrieval excludes Prosecutor evidence	Shared vs. disjoint evidence comparison
A real case may not support the claim [6, 12]	Separate case existence from holding support	Hard-negative stress tests, expert labels
Agreement and confidence can mislead [5, 6]	Citation-aware controller; Judge limited to verified list	Calibration and abstention
No system remembers rejected citations [13, 27]	Verdict cache + proposition-level risk memory	Paraphrase tests, cache isolation

Introduction & Problem Recap

Problem and Objective



58–88%

Hallucination rate of public LLMs on direct, verifiable questions about U.S. federal cases

GPT-4 58% · GPT-3.5 69% · PaLM 2 72% · Llama 2 88% —
Dahl et al., 2024 [6]

PROBLEM

Legal answers can contain invented cases, wrong citation identities and holdings that a real case does not support.

OBJECTIVE

Combine legal retrieval, adversarial debate and citation verification to reduce unsupported legal answers.

CURRENT SCOPE

U.S. federal case law from Harvard CAP. Prosecutor, Defense, Reflection and Judge share one 4-bit Mistral-7B on a single T4 GPU.

Introduction & Problem Recap (contd.)

Objectives and Research Questions



#	OBJECTIVE	RESEARCH QUESTION
O1	Ground answers in legal sources	Does hybrid retrieval recover the target authority and the passage supporting its holding?
O2	Create an evidence-diverse debate	Does independent Defense retrieval reveal errors that shared-context debate misses?
O3	Verify each cited proposition	Can the verifier separate existence, support, fabrication and missing corpus coverage?
O4	Control debate effort	Does dynamic stopping improve quality per unit of latency over a fixed-round baseline?
O5	Evaluate reproducibly	Do gains persist on held-out cases with a fixed corpus, isolated memory and checked labels?

Introduction & Problem Recap (contd.)

Improvements and Further Proposed



AT REVIEW-I

Weighted-sum citation score

Shared-context Defense

Hand-picked 0.40 / 0.75 cut-offs

Name-level risk list

End-to-end RAGAS only

REFINEMENT FOR REVIEW-II

Factorized CCS = Existence × Support with a five-way verdict

Evidence-disjoint Defense retrieves its own counter-evidence

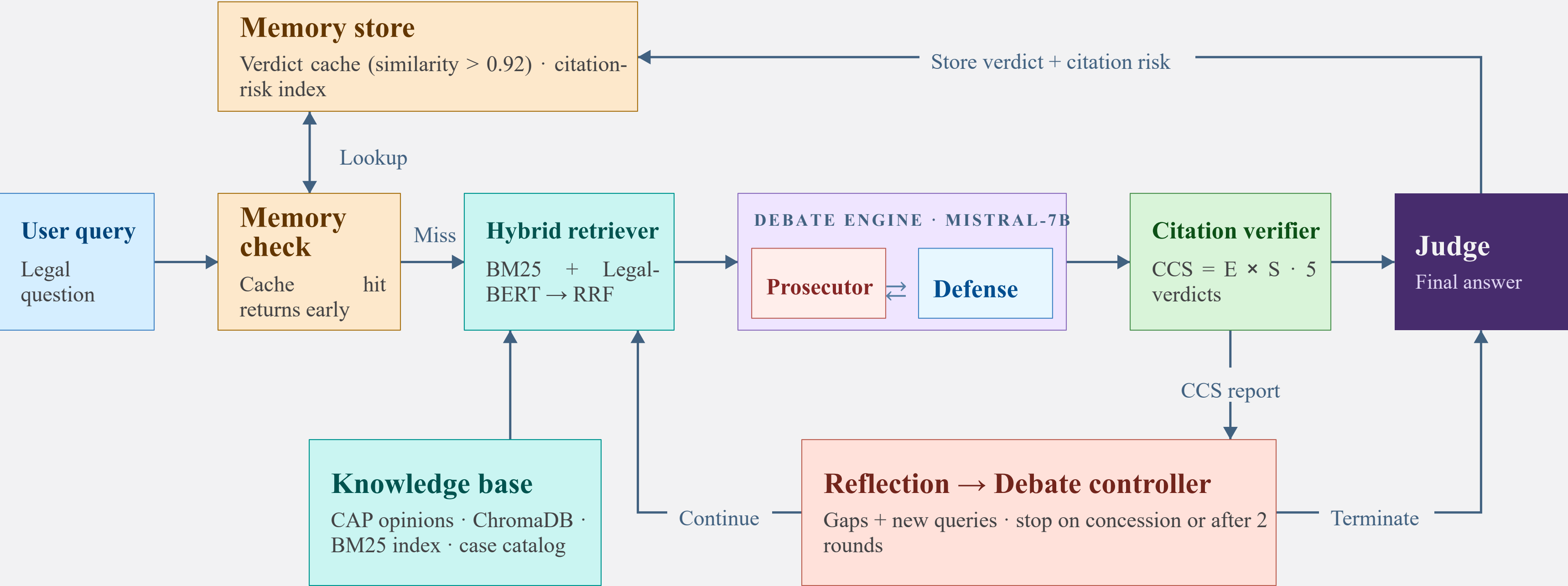
Based on CCS

Self-correcting memory of (case, wrong holding) pairs

40-citation stress test and four-variant ablation study

System Design & Architecture

System Architecture



System Design & Architecture (contd.)

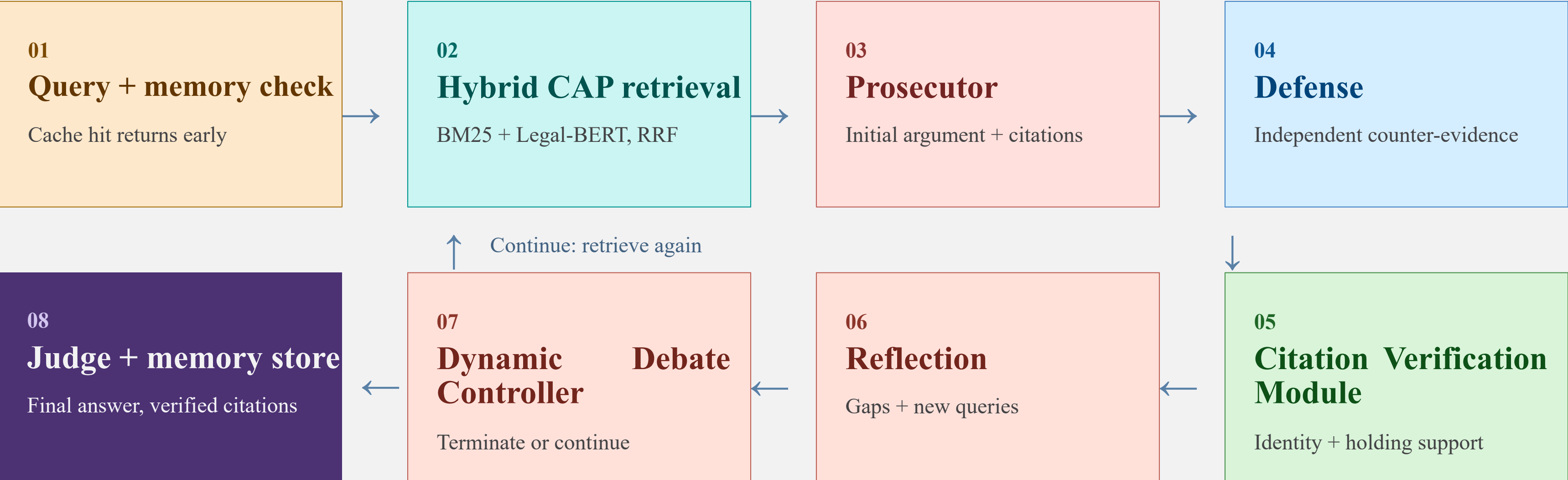
System Components



DATA	CAP case opinions 27 U.S. Reports volumes	Case catalog 14,524 name / reporter keys	Passage store 450-word chunks + metadata	Memory store Verdict cache + risk index
RETRIEVAL	BM25 lexical Names + terms · top 4	Legal-BERT dense ChromaDB · top 4	RRF fusion $k = 60 \cdot \leq 8$ passages	Defense counter-retrieval Prosecutor docs excluded
AGENTS	Prosecutor Argument + citations	Defense Challenges + concession	Reflection Gaps + new queries	Judge Verified citations only
VERIFICATION	Existence · E Catalog + fuzzy name	Support · S DeBERTa NLI entailment	CCS = E × S Per-citation score	Five-way verdict Supported → Unverifiable
CONTROL	LangGraph state Query, evidence, history	Debate controller Stop or continue · ≤ 2 rounds	Mistral-7B · NF4 Shared by all roles · T4 GPU	Evaluation runners Benchmark + ablation

System Design & Architecture (contd.)

Debate Workflow



ORCHESTRATION

LangGraph state machine

DEBATE BUDGET

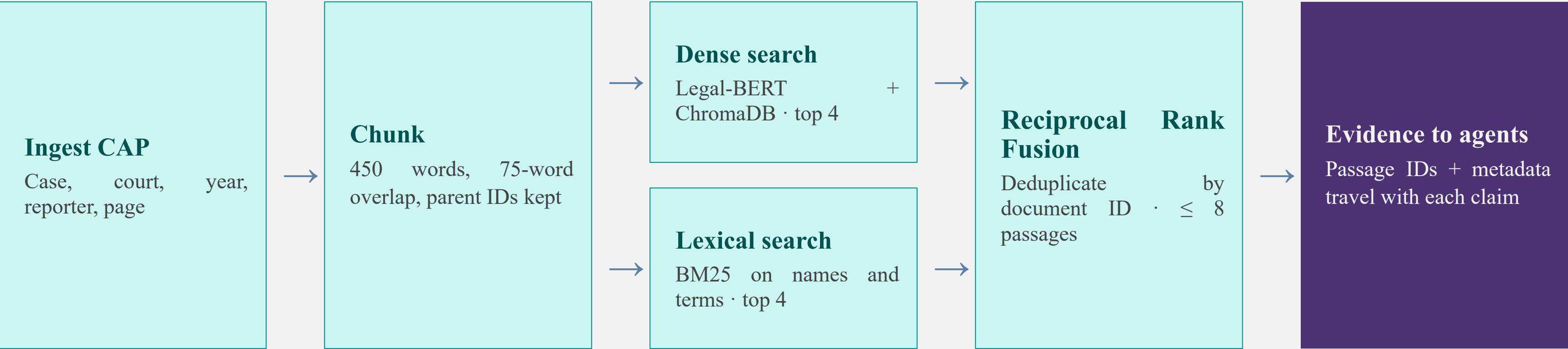
Maximum two rounds per query

PROVENANCE

Passage IDs kept through debate and verification

System Design & Architecture (contd.)

Hybrid Retrieval Pipeline



$$\text{RRF}(d) = \sum 1 / (60 + \text{rank}_i(d))$$

Exact case names and reporter keys come from BM25; paraphrased holdings come from dense search. Fusion rewards passages that rank well in both.

System Design & Architecture (contd.)

Agent Roles and Evidence-Disjoint Defense



Prosecutor Drafts an answer from retrieved case passages Argument + candidate citations	Defense Retrieves its own evidence and challenges the claims Challenges, counter-citations, concession	Reflection Reviews the debate and citation-confidence feedback Gaps, targeted queries, quality score	Judge Synthesises the debate; keeps only CVM-supported citations Answer, verdict, confidence
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Defense independence	COUNTER-QUERIES Question + exception terms; cited case + “overruled”, “distinguished”	EVIDENCE EXCLUSION Drop passages matching Prosecutor document IDs; keep up to four	EVIDENCE OVERLAP RATIO Share of Defense citations found only in Prosecutor evidence. 1.0 = echo
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System Design & Architecture (contd.)

Factorized Citation Verification



$$CCS = E \times S$$

E · Existence

Exact catalog match, reporter/name check, fuzzy name match

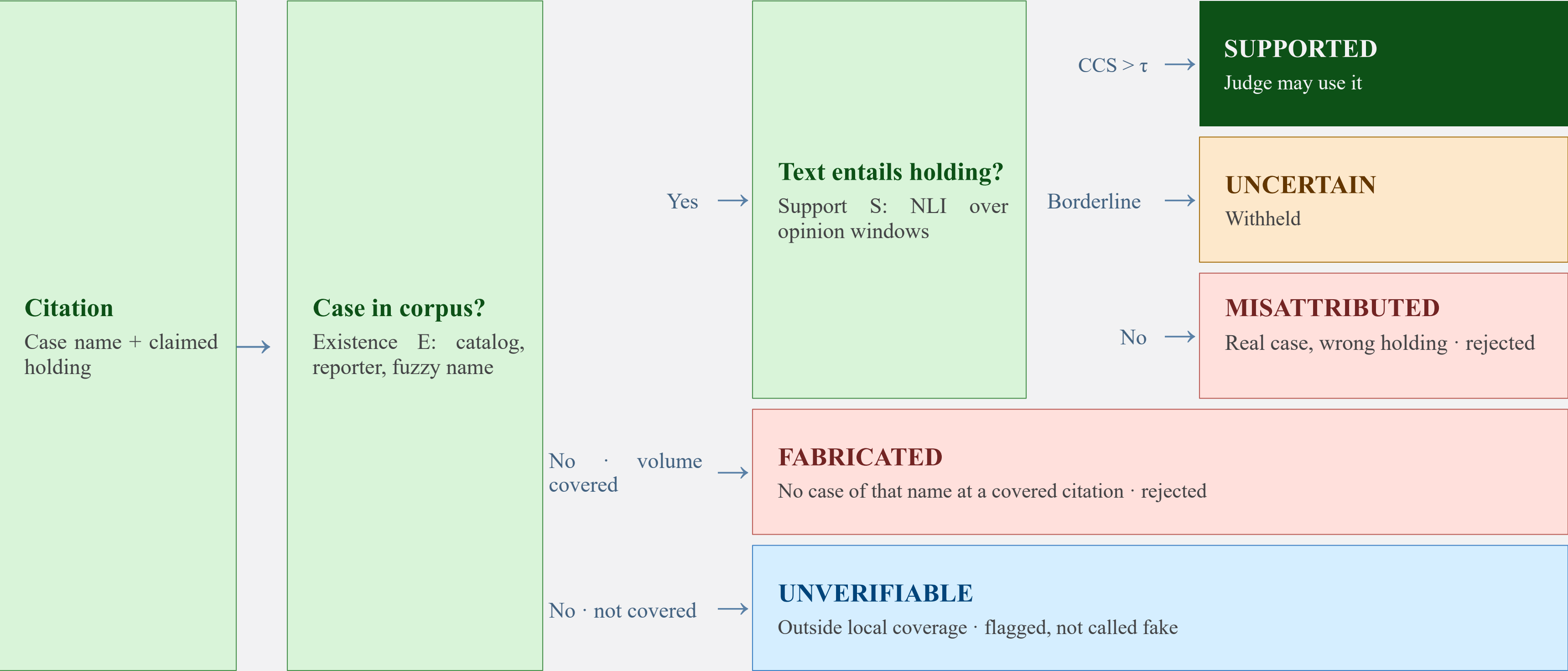
S · Holding support

DeBERTa NLI cross-encoder: does the case's own text entail the claimed holding?

VERDICT	CONDITION	JUDGE MAY USE?
SUPPORTED	$E \geq 0.85$ and $S \geq 0.70$	Yes
UNCERTAIN	$E \geq 0.85$ and $0.40 \leq S < 0.70$	No
MISATTRIBUTED	$E \geq 0.85$ and $S < 0.40$	Rejected with reason
FABRICATED	Low E in a covered reporter volume	Rejected with reason
UNVERIFIABLE	Outside local corpus coverage	Flagged, not called fake

System Design & Architecture (contd.)

Citation Verdict Decision Flow



System Design & Architecture (contd.)



Proposed Novel Contributions

01

Factorized citation verification

CCS = Existence $\times$ Support with a five-way verdict

Replaces the weighted CVM sum

02

Evidence-disjoint Defense

Defense retrieves its own counter-evidence;
Prosecutor documents excluded

Replaces shared-context Defense

03

Citation stress-test protocol

Inject labelled bad citations; measure catch rate per error type

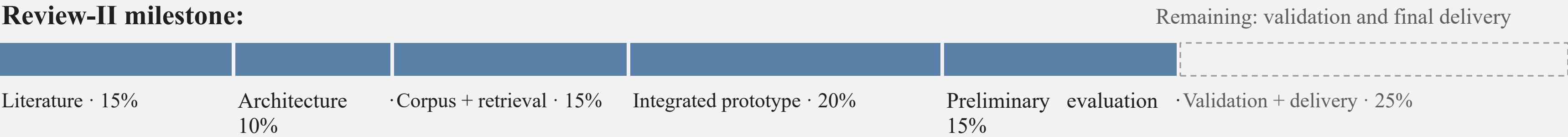
Replaces end-to-end RAGAS only

CONTRIBUTION STATEMENT

LexAgent distinguishes fabricated, misattributed and unverifiable citations, forces evidential independence between debating agents, and bounds how often unsupported citations are accepted, all on one T4 GPU.

Implementation Details

Modules Developed and Integrated · Progress



LAYER	TECHNOLOGY	CONFIGURATION
Generation	Mistral-7B-Instruct-v0.3, NF4 4-bit	Tesla T4 16 GB · 768 max tokens · temp 0.1
Retrieval	Legal-BERT + ChromaDB, BM25, RRF (k = 60)	Top 4 + 4 → 8 passages · 450-word chunks · 27 volumes
Verification	DeBERTa NLI cross-encoder, factorized CVM	Per-citation identity and support reports
Orchestration	LangGraph, four roles, DDC, adaptive memory	Max two rounds · cache reuse at similarity > 0.92
Evaluation	Python CLI, Colab notebook, benchmark + ablation runners	25-case scorecard · 40-item stress test

Implementation Details (contd.)

Algorithms Implemented

Reciprocal Rank Fusion

$\text{score}(d) = \sum 1 / (k + \text{rank})$, $k = 60$; BM25 top 4 + Legal-BERT top 4
→ 8 passages

Factorized citation score

$\text{CCS} = E \times S$; verdict from existence and holding-support thresholds

NLI holding support

DeBERTa cross-encoder tests whether the case text entails the claimed holding

Evidence exclusion

Defense drops passages matching Prosecutor document IDs; keeps up to four

Dynamic Debate Controller

Terminate or continue after Reflection; maximum two rounds

Adaptive memory

Reuse cached answers at similarity > 0.92 ; store Judge verdicts



Results & Analysis (75%)

Benchmark Results



25/25

Cases completed, no crashes

12 / 25

Cases with HR = 0

91.5%

Mean citation accuracy (CAS)

0.000

Evidence overlap ratio

72%

Debate convergence (18/25)

73.2 min

One T4 GPU · ≈ 2.9 min per case

19.4%

Mean hallucination rate (HR)

20 / 5

Two-round / one-round debates

READING THE HR

HR includes authentic Supreme Court citations outside the 27-volume local preset, such as Lockett v. Ohio and Milliken v. Bradley. These are unverifiable by this corpus, not confirmed fabrications.

Results & Analysis (contd.)

Citation Stress Test



CATEGORY	EXPECTED	ACCEPTED BY VERIFIER		OUTCOME
Correct precedents (control)	Accept		10/10	As expected
Semantic misattributions	Reject		10/10	None caught
CaseHOLD legal distractors	Reject		5/5	Lexical overlap passes
Plausible fabricated citations	Reject		1/5	4 of 5 rejected
Fabricated, no citation details	Reject		0/5	All rejected
Real, off-corpus precedents	Flag		0/5	Unverifiable (coverage)

100%

In-corpus control acceptance (10/10)

90%

Fabrication catch rate (9/10)

0%

Semantic misattribution catch (0/10)

Results & Analysis (contd.)

Component Ablation Study



VARIANT	MEAN HR	CAS	CONFID.
Full · DVA+ Complete LexAgent v3.0	0.168	0.911	0.763
A1 · No Defense Monologue baseline	0.000	0.906	0.766
A2 · No CVM Citation verification removed	0.133	0.914	0.700
A3 · No memory Adaptive memory removed	0.225	0.926	0.865
A4 · No DDC Fixed one round	0.058	0.909	0.780

+33.9%

Relative HR increase without
adaptive memory (0.168 → 0.225)

CAUTION

A1's zero HR comes from removing the
adversary, not from stronger robustness.

CONFIDENCE

A3 has the highest HR and the highest
confidence, so confidence alone is not
reliability.

Results & Analysis (contd.)

Worked Example



QUERY

Does the Fourth Amendment protect against warrantless cell phone searches?

PHASE	SAVED OBSERVATION	WHAT IT SHOWS
1 · Hybrid retrieval	8 CAP authority passages returned	Dense and lexical retrieval work together
2 · Fixed one-round debate	Riley citation rejected as unverified; Judge confidence 0.35	An answer can outlive a failed citation check
3 · Full DVA+	2 rounds, average CCS 0.998, one verified citation, confidence 1.00	Verification and stopping state are recorded

The separate benchmark run reports 66.67% HR for this case, so versions will be pinned and trials repeated.

Results & Analysis (contd.)

Summary of Findings



Prototype achieved

CAP retrieval, four-role debate, factorized citation verification, dynamic stopping and adaptive memory run end-to-end on one T4.

Open issues

Semantic misattributions and CaseHOLD distractors pass the verifier; off-corpus citations inflate the measured HR.

Next milestone

Repair holding-support checks, calibrate τ conformally, then test against a matched RAG baseline and ablations.

Challenges & Remaining Work

Challenges Faced and Solutions



CHALLENGE	SOLUTION / STATUS
16 GB T4 memory for four agents	One shared Mistral-7B in NF4 4-bit; roles differ by prompt and evidence
Defense echoing Prosecutor evidence	Evidence-disjoint retrieval; evidence overlap ratio 0.000 on the benchmark
Authentic off-corpus citations counted in HR	Coverage-aware UNVERIFIABLE verdict; HR reported with this caveat
Semantic misattributions pass the verifier (0/10)	Open: holding-support repair planned before Review-III
Heuristic acceptance thresholds	Conformal calibration of τ planned
Run-to-run variation between notebook and benchmark	Pin model and library versions; repeat trials
Latency: 73.2 min for 25 cases	Two-round debate cap and memory cache reuse

Challenges & Remaining Work (contd.)

Remaining Work (25%) and Timeline for Completion by Review-III (28.10.2026)



10%

Verifier + final-answer checks

8%

Baselines + ablations

4%

Memory + reproducibility

3%

Final report + demo

30 Sep – 6 Oct

Audit identity resolution, NLI mapping, final-answer checks

Verifier tests, revised stress test

7 – 13 Oct

Matched RAG baseline and four ablations

Comparable scorecards, latency, tokens

14 – 20 Oct

Held-out review, paraphrase and memory tests

Independent labels, failure taxonomy

21 – 27 Oct

Repeat experiments; finalise report and demo

Reproducible package, final figures

28 Oct

Review-III

Final results and limitations

Conclusion



LexAgent v3.0 presents a citation-grounded, multi-agent debate architecture for reducing hallucinations in legal question answering under strict computational and privacy constraints. The system integrates Harvard’s Caselaw Access Project (CAP) as its authority database, a hybrid retrieval pipeline combining BM25 and Legal-BERT through Reciprocal Rank Fusion (RRF), and an adversarial Prosecutor–Defense–Judge protocol orchestrated using LangGraph. Together, these components introduce structured evidential scrutiny before the final synthesis of legal answers.

At the Review-II milestone, representing 75% of the scheduled development, experimental evaluation supports three principal findings.

Evidential Independence Reduces False Consensus.

Requiring the Defense agent to conduct independent counter-retrieval while excluding passages retrieved by the Prosecutor produced an Evidence Overlap Ratio (EOR) of 0.000 across all 25 landmark benchmark cases. This separation prevents the Defense agent from simply reproducing the Prosecutor's evidence and directly addresses the echo-chamber problem associated with multi-agent debate systems.

Factorized Verification Prevents Identity–Holding Masking.

Replacing the previous additive confidence heuristic with the factorized formulation $CCS = E \times \text{Sensures}$ that recognition of an authentic case identity cannot compensate for an unsupported or fabricated holding. The accompanying five-way verdict taxonomy further distinguishes authentic but off-corpus authorities (UNVERIFIABLE) from genuinely fabricated claims (FABRICATED), providing a more granular mechanism for citation verification.

Cross-Session Memory Provides Structural Risk Control.

Component ablation demonstrates the importance of proposition-level adaptive memory. Removing this mechanism increased the hallucination rate by 33.9%, from 0.168 to 0.225, while simultaneously producing the highest ungrounded model confidence (0.865). This divergence between confidence and evidential reliability indicates that persistent risk tracking is important for stabilizing stochastic LLM behavior across sessions.

The remaining 25% of development toward Review-III will focus on two primary objectives. First, the conformal acceptance threshold τ will be formally calibrated to establish statistical error guarantees of the form

$$P(\text{accept} \mid \text{unsupported}) \leq \alpha.$$

Second, the system will address the sensitivity of NLI-based cross-encoders to lexical overlap, particularly when distinguishing legally relevant propositions from fine-grained legal distractors.

Overall, the Review-II results indicate that LexAgent's combination of independent evidence retrieval, factorized citation verification, adversarial deliberation, and persistent proposition-level memory provides a structured foundation for citation-grounded legal question answering. The final development stage will therefore shift from architectural validation toward statistical calibration and robustness against difficult retrieval and entailment cases.

References

[1]–[15]



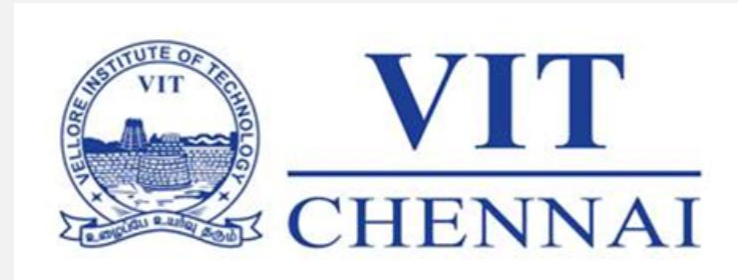
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Thank You

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